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PHARMACY DATABASE SYSTEM

This project was developed as a group assignment, with my contribution in selected parts of the implementation.

AUTHOR

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1. Database Design (ERD)

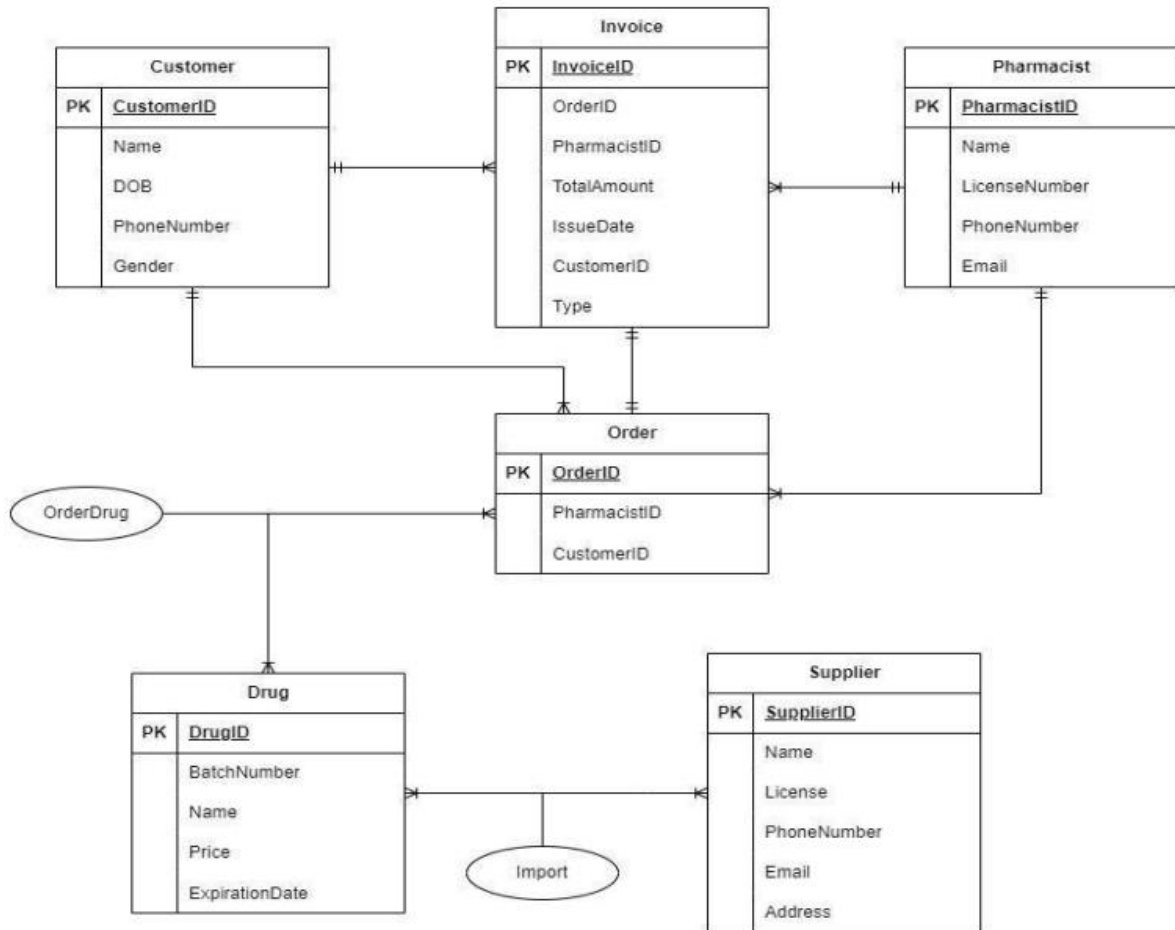


Figure 1: ERD

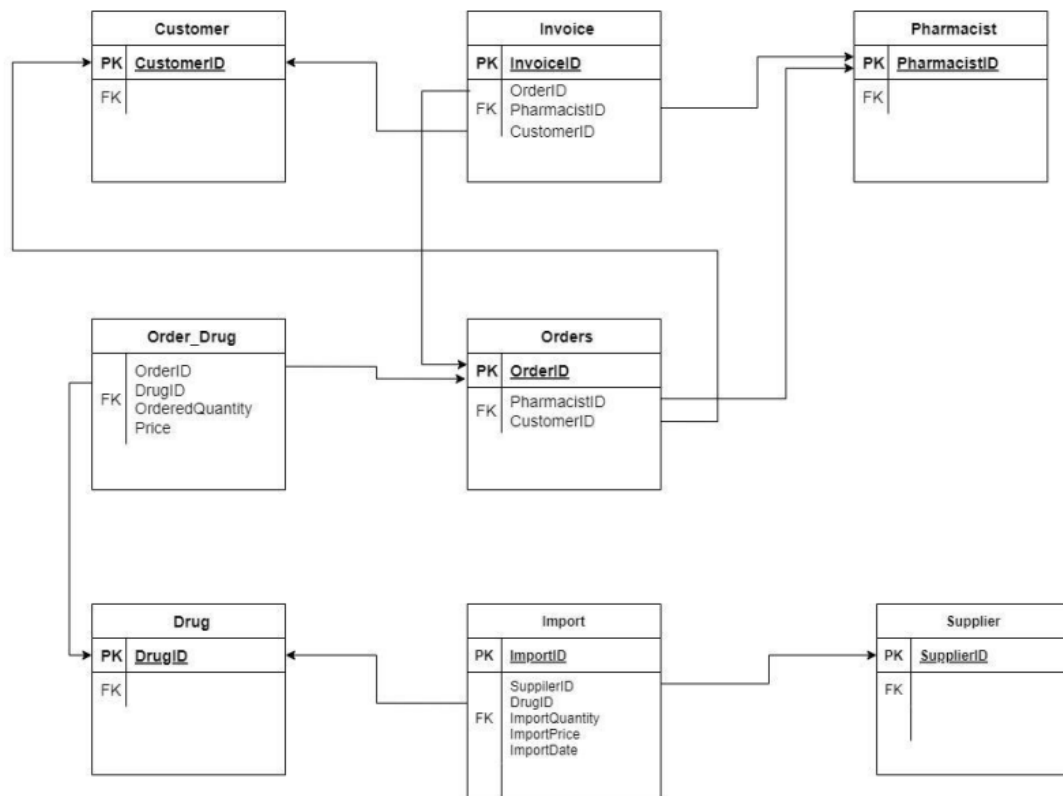


Figure 2: Relation Schema

2. Database Structure (Simplified)

Below are selected tables used in the system:

```
CREATE TABLE Customer (
    CustomerID INT NOT NULL,
    Name NVARCHAR(100),
    PhoneNumber NVARCHAR(20),
    Gender NVARCHAR(10),
    PRIMARY KEY (CustomerID)
);
```

```
CREATE TABLE Drug (  
    DrugID NVARCHAR(20) NOT NULL,  
    BatchNumber NVARCHAR(10),  
    Name NVARCHAR(100),  
    Price INT,  
    StockQuantity INT,  
    Unit NVARCHAR(10),  
    ExpirationDate NVARCHAR(20),  
    PRIMARY KEY (DrugID)  
);
```

```
CREATE TABLE [Order] (  
    OrderID INT NOT NULL,  
    CustomerID INT,  
    PRIMARY KEY (OrderID),  
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)  
);
```

```
CREATE TABLE Invoice (  
    InvoiceID NVARCHAR(20) NOT NULL,  
    OrderID INT,  
    TotalAmount INT,  
    IssueDate NVARCHAR(20),
```

```
CustomerID INT,  
  
PRIMARY KEY (InvoiceID),  
  
FOREIGN KEY (OrderID) REFERENCES [Order](OrderID)  
  
);
```

3. Selected SQL Queries

3.1 Delete Old Invoices

```
DELETE FROM Invoice  
WHERE IssueDate < DATEADD(YEAR, -2, GETDATE());
```

Messages

(3 rows affected)

Completion time: 2024-06-17T23:55:14.6477757+07:00

This query removes invoice records older than 2 years from the system.

Purpose:

- Helps maintain database performance by reducing unnecessary data
- Ensures the system only stores relevant and recent transaction records.

3.2 Check Drug Availability

```
SELECT  
    Name AS DrugName,  
    StockQuantity,  
    CASE  
        WHEN StockQuantity >= 3 THEN 'Available'  
        ELSE 'Not Available'  
    END AS Availability  
FROM Drug  
WHERE DrugID = '761778205867';
```

Results Messages			
	DrugName	StockQuantity	Availability
1	COLON CARE	95	Available

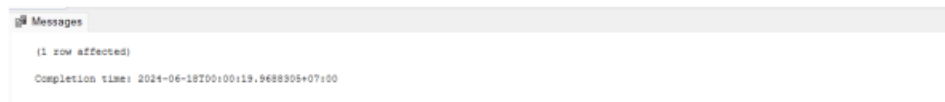
This query checks whether a specific drug is available in stock based on its quantity.

Purpose:

- Supports order processing by verifying stock availability before confirming a purchase
- Helps prevent situations where orders are placed for out-of-stock products

3.3 Insert New Drug with Duplicate Check

```
INSERT INTO Drug (DrugID, BatchNumber, Name, Price, StockQuantity, Unit,
ExpirationDate)
SELECT '5729480631', '875621' , 'FALGANKID 160', 3465, 88, 'Tube', '2028-07-02'
WHERE NOT EXISTS (
    SELECT 1
    FROM Drug
    WHERE DrugID = '5729480631'
);
```



This query inserts a new drug into the database only if the DrugID does not already exist.

Purpose:

- Prevents duplicate records in the system
- Ensures data integrity during the insertion process

4. Summary

This project demonstrates basic database design and SQL usage in a pharmacy management scenario.

My contributions focused on implementing SQL queries for selected business cases, including data validation, availability checking, and controlled data insertion. Through this project, I gained hands-on experience in working with relational data, understanding system workflows, and applying SQL to support simple business logic.